

As there's only Analog read functionality, you need to explicitly specify the pin as INPUT or OUTPUT.

Code:

```
• void setup(){
•   pinMode(A0, INPUT);
• }
• void loop(){
•   analogRead(A0);
• }
```

Analog Write / PWM

The 'analog write' function is a digital form of the Analog signal. The signal is commonly referred to as the PWM signal. This function can only be used on Pins 3, 5, 6, 9, 10 and 11 on the Arduino™ boards.

The syntax for using the analog write function is:

```
• analogWrite (pin, value)
```

The range of value for the analog write function is 0 to 255, where 0 corresponds to zero voltage and 255 corresponds to maximum voltage i.e. +5 V. This value is the PWM form, thus the average voltage obtained at the output depends on the term called the 'duty cycle' of the wave.

Code:

```
• int led = 10;
• void setup () { }
• void loop (){
•   analogWrite (led, 127); // Generates a PWM wave of 50%
duty cycle of average voltage of 2.5 V.
•   delay (5000);
•   analogWrite (led, 64); // Generates a PWM wave of 25%
duty cycle of average voltage of 1.25 V.
•   delay (5000);
• }
```

Delay

The 'delay' function causes the execution of the Arduino™ board to pause for the specified amount of time.

```
• delay(1000); // wait for 1 second
```

The parameter passed to the delay function is the time that it should